

Notes on Introduction to Real Analysis

1 Multivariable Differential Calculus

1.1 Linear Approximation

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be defined by $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$, where each component $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued function.

Regardless of whether f is linear or non-linear, a natural question arises: can it be locally approximated by a linear function? And if so, how accurate is this approximation?

Ultimately, given a point $\mathbf{x}_0 \in \mathbb{R}^n$, our goal is to approximate f near $\mathbf{x}_0$ using its linearization:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + Df(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$$

Definition 1.1 (Differentiability)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We say f is differentiable at $\mathbf{x}_0$ with derivative L if

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0.$$

Equivalently,

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - L(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Therefore, multivariable differentiability is defined such that the relative error between the function and its linear approximation vanishes as the distance approaches zero.

Theorem 1.1

Let $E \subseteq \mathbb{R}$, $f : E \rightarrow \mathbb{R}$ be a function, $L \in \mathbb{R}$, and x_0 be an interior point of E . TFAE:

1. f is differentiable at x_0 and $f'(x_0) = L$.
2. $\lim_{x \rightarrow x_0} \frac{|f(x) - f(x_0) - L(x - x_0)|}{|x - x_0|} = 0$.

Proof.

For $x \neq x_0$, write

$$\frac{f(x) - f(x_0)}{x - x_0} = L + \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0}.$$

Rearrange L to the left, take the absolute value and limit $x \rightarrow x_0$, and we can show the limit on the both sides imply each other. The second expression is exactly the special case of definition (1.1). ■

Theorem 1.2 (Uniqueness of derivatives)

Let $\mathbf{x}_0 \in E$ be an interior point of E . Suppose $L_1, L_2 : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear transformations such that f is differentiable at $\mathbf{x}_0$ with derivative L_1 and also differentiable at $\mathbf{x}_0$ with derivative L_2 . Then $L_1 = L_2$.

Proof.

For $\mathbf{h} \neq \mathbf{0}$,

$$\frac{\|L_1(\mathbf{h}) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|} \leq \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - L_1(\mathbf{h})\|}{\|\mathbf{h}\|} + \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|}.$$

By sandwich, $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|L_1(\mathbf{h}) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|} = 0$.

Let $\mathbf{h} = t\mathbf{v}$, $t \neq 0$, we have

$$\lim_{\mathbf{v} \rightarrow \mathbf{0}} \frac{\|L_1(t\mathbf{v}) - L_2(t\mathbf{v})\|}{\|t\mathbf{v}\|} = \lim_{\mathbf{v} \rightarrow \mathbf{0}} \frac{|t| \|L_1(\mathbf{v}) - L_2(\mathbf{v})\|}{|t| \|\mathbf{v}\|} \quad (L_1 \text{ and } L_2 \text{ are linear.})$$

Hence, for $\mathbf{v} \neq \mathbf{0}$, $L_1(\mathbf{v}) = L_2(\mathbf{v})$, which is also true when $\mathbf{v} = \mathbf{0}$. ■

We may prove by contradiction that f is not differentiable by the above theorem.

Theorem 1.3

If f is differentiable at $\mathbf{x}_0$, then f is continuous at $\mathbf{x}_0$.

Proof.

For $\mathbf{x} \neq \mathbf{x}_0$, write

$$\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\| = \underbrace{\frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|}}_{\rightarrow 0 \text{ by assumption}} \underbrace{\|\mathbf{x} - \mathbf{x}_0\|}_{\rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{x}_0}.$$

Next, use

$$\|f(\mathbf{x}) - f(\mathbf{x}_0)\| \leq \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}_{\rightarrow 0 \text{ from the previous equality}} + \underbrace{\|L(\mathbf{x} - \mathbf{x}_0)\|}_{\rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{x}_0}.$$

By Sandwich, $\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$. ■

Definition 1.2 (Directional Derivative)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Fix $\mathbf{v} \in \mathbb{R}^n$, if the limit

$$D_{\mathbf{v}}f(\mathbf{x}_0) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t}$$

exists, we call this limit the directional derivative of f at $\mathbf{x}_0$ in the direction $\mathbf{v}$.

Definition 1.3 (Partial Derivative)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Let $1 \leq j \leq n$, if the limit

$$\frac{\partial f}{\partial x_j}(\mathbf{x}_0) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{e}_j) - f(\mathbf{x}_0)}{t}$$

exists, we call this limit the partial derivative of f with respect to the variable x_j .

So partial derivative is just a special case of directional derivative.

Theorem 1.4

If f is differentiable at $\mathbf{x}_0$, then f is differentiable in the direction $\mathbf{v}$ at $\mathbf{x}_0$, and:

$$D_{\mathbf{v}}f(\mathbf{x}_0) = f'(\mathbf{x}_0) \cdot \mathbf{v}.$$

Proof.

For $\mathbf{h} \neq 0$, let $\mathbf{h} = t\mathbf{v}$, $t \neq 0$, write

$$\frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t} = \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0) - L(t\mathbf{v})}{t} + \frac{L(t\mathbf{v})}{t}.$$

Rearrange $L(\mathbf{v})$ to the left, and take the Euclidean norm and $t \rightarrow 0$, the RHS converge to 0 by differentiability ($\|\mathbf{v}\| \neq 0$). So we have

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t} = L(\mathbf{v}) = f'(\mathbf{x}_0) \cdot \mathbf{v}.$$

■

We may also disprove the differentiability of f provided its directional derivative exists. The next question is: how to write the linear map $f'(\mathbf{x}_0)$ explicitly?

Theorem 1.5

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . If all the partial derivatives of f exist on E and are continuous at $\mathbf{x}_0$, then f is differentiable at $\mathbf{x}_0$, and

$$f'(\mathbf{x}_0) \cdot \mathbf{v} = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0).$$

Proof.

Step 1: Setup

Let $L(\mathbf{v}) = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0)$ be our candidate for the derivative. Since $\mathbf{x}_0$ is an interior point of E , there exists $r > 0$ such that the ball $B(\mathbf{x}_0, r) \subseteq E$.

For $\epsilon > 0$, since each partial derivative $\frac{\partial f_i}{\partial x_j}$ is continuous at $\mathbf{x}_0$, there exists $\delta_{i,j} > 0$ such that $\|\mathbf{x} - \mathbf{x}_0\| < \delta_{i,j}$ implies

$$\left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| < \frac{\epsilon}{mn}$$

for all $1 \leq i \leq m$ and $1 \leq j \leq n$.

Let $\delta = \min\{\delta_{i,j} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$. Now, fix $\mathbf{x} \in E$ such that $\|\mathbf{x} - \mathbf{x}_0\| < \delta$.

Let $\mathbf{v} = \mathbf{x} - \mathbf{x}_0 = \sum_{j=1}^n v_j \mathbf{e}_j$. Note that $|v_j| \leq \|\mathbf{x} - \mathbf{x}_0\| < \delta$. We construct a path from $\mathbf{x}_0$ to $\mathbf{x}$ along the coordinate axes. Define the sequence of points:

$$\begin{aligned} \mathbf{x}^{(0)} &= \mathbf{x}_0 \\ \mathbf{x}^{(1)} &= \mathbf{x}_0 + v_1 \mathbf{e}_1 \\ \mathbf{x}^{(2)} &= \mathbf{x}_0 + v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 \\ &\vdots \\ \mathbf{x}^{(j)} &= \mathbf{x}^{(j-1)} + v_j \mathbf{e}_j \\ &\vdots \\ \mathbf{x}^{(n)} &= \mathbf{x}_0 + \sum_{j=1}^n v_j \mathbf{e}_j = \mathbf{x} \end{aligned}$$

We can express the difference $f(\mathbf{x}) - f(\mathbf{x}_0)$ as a telescoping sum:

$$f(\mathbf{x}) - f(\mathbf{x}_0) = \sum_{j=1}^n \left(f(\mathbf{x}^{(j)}) - f(\mathbf{x}^{(j-1)}) \right)$$

For each component i , consider $f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)})$. Let $\varphi_i^j(t) = f_i(\mathbf{x}^{(j-1)} + t v_j \mathbf{e}_j)$, then $f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) = \varphi_i^j(1) - \varphi_i^j(0)$. By the Mean Value Theorem, $\exists t_j \in (0, 1)$ such that $\varphi_i^j(1) - \varphi_i^j(0) = (\varphi_i^j)'(t_j)$.

Step 2: Continuity of f_i along the coordinate $\mathbf{x}_j$

Applying Lemma (1.1):

$$f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) = (\varphi_i^j)'(t_j) = \frac{\partial f_i}{\partial x_j}(\xi_j) v_j,$$

where $\xi_j = \mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j$.

Note that

$$\begin{aligned}
\|\mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j - \mathbf{x}_0\| &= \|v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_{j-1} \mathbf{e}_{j-1} + t_j v_j \mathbf{e}_j\| \\
&\leq \left\| \sum_{i=1}^n v_i \mathbf{e}_i \right\| \quad (\because |t_j| < 1) \\
&= \|\mathbf{x} - \mathbf{x}_0\| \leq \delta,
\end{aligned}$$

which implies

$$\begin{aligned}
\left| f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| &= \left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| |v_j| \\
&< \frac{\epsilon}{mn} |v_j|.
\end{aligned}$$

Step 3: Bound the error for each component f_i

$$\begin{aligned}
\left| f_i(\mathbf{x}) - f_i(\mathbf{x}_0) - \sum_{j=1}^n \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| &= \left| \sum_{j=1}^n \left(f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right) \right| \\
&\leq \sum_{j=1}^n \left| \frac{\partial f_i}{\partial x_j}(\boldsymbol{\xi}_j) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| |v_j| \\
&< \sum_{j=1}^n \frac{\epsilon}{mn} |v_j| \\
&\leq \sum_{j=1}^n \frac{\epsilon}{mn} \|\mathbf{x} - \mathbf{x}_0\| = \frac{\epsilon}{m} \|\mathbf{x} - \mathbf{x}_0\|
\end{aligned}$$

Step 4: Extend this to the vector-valued function f

$$\begin{aligned}
\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\| &\leq \sum_{i=1}^m \left| f_i(\mathbf{x}) - f_i(\mathbf{x}_0) - \sum_{j=1}^n \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| \\
&< \sum_{i=1}^m \frac{\epsilon}{m} \|\mathbf{x} - \mathbf{x}_0\| = \epsilon \|\mathbf{x} - \mathbf{x}_0\|
\end{aligned}$$

Dividing both sides by $\|\mathbf{x} - \mathbf{x}_0\|$, we get:

$$\frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} < \epsilon$$

Since this holds for any $\epsilon > 0$ when $\|\mathbf{x} - \mathbf{x}_0\|$ is sufficiently small, we conclude:

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0$$

This proves that f is differentiable at $\mathbf{x}_0$, and its derivative is given by L . ■

Lemma (Chain rule along a coordinate segment)

Let $E \subset \mathbb{R}^n$ and $f : E \rightarrow \mathbb{R}^m$. Fix $1 \leq i \leq m$ and $1 \leq j \leq n$. Define

$$\varphi_i^j(t) := f_i(\mathbf{z} + tv\mathbf{e}_j), \quad 0 \leq t \leq 1,$$

where $\mathbf{z} \in E$ and $v \in \mathbb{R}$. Assume that for some $t_0 \in (0, 1)$ there exists $\rho > 0$ such that

$$\mathbf{z} + (t_0 + h)v\mathbf{e}_j \in E \quad \text{for all } |h| < \rho,$$

and that the partial derivative $\frac{\partial f_i}{\partial x_j}$ exists at the point $\mathbf{p} := \mathbf{z} + t_0v\mathbf{e}_j$. Then φ_i^j is differentiable at t_0 and

$$(\varphi_i^j)'(t_0) = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v.$$

Proof.

Let

$$\begin{aligned} (\varphi_i^j)'(t_0) &= \lim_{h \rightarrow 0} \frac{\varphi_i^j(t_0 + h) - \varphi_i^j(t_0)}{h} = \lim_{h \rightarrow 0} \frac{f_i(\mathbf{z} + (t_0 + h)v\mathbf{e}_j) - f_i(\mathbf{z} + t_0v\mathbf{e}_j)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f_i(\mathbf{p} + hv\mathbf{e}_j) - f_i(\mathbf{p})}{h}. \end{aligned}$$

If $v = 0$, $f_i(\mathbf{p} + hv\mathbf{e}_j) - f_i(\mathbf{p}) = 0$. Now consider $v \neq 0$.

For arbitrarily small h , $\mathbf{z} + (t_0 + h)v\mathbf{e}_j = \mathbf{p} + hv\mathbf{e}_j \in B(\mathbf{p}, \rho) \subseteq E$. Set $hv = s$, the above quotient becomes:

$$\lim_{s \rightarrow 0} \frac{f_i(\mathbf{p} + s\mathbf{e}_j) - f_i(\mathbf{p})}{s} \cdot v = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v,$$

which shows that φ_i^j is differentiable at t_0 and $(\varphi_i^j)'(t_0) = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v$. ■

Remark. Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . If f is differentiable at $\mathbf{x}_0$, by definition, there exists a linear map such that the relative error converge to zero near $\mathbf{x}_0$. Then we call this linear map (total) derivative of f at $\mathbf{x}_0$, denoted by

$$Df(\mathbf{x}_0) = f'(\mathbf{x}_0) = \begin{pmatrix} - & \nabla f_1(\mathbf{x}_0) & - \\ - & \nabla f_2(\mathbf{x}_0) & - \\ & \vdots & \\ - & \nabla f_m(\mathbf{x}_0) & - \end{pmatrix}_{m \times n}$$

$$= \begin{pmatrix} \left| \frac{\partial f}{\partial x_1}(\mathbf{x}_0) \right| & \left| \frac{\partial f}{\partial x_2}(\mathbf{x}_0) \right| & \cdots & \left| \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \right| \\ \left| \right| & \left| \right| & & \left| \right| \end{pmatrix}_{m \times n}.$$

Hence, we have:

1. The partial derivative is simply the directional derivative along the standard basis vector $\mathbf{e}_j$:

$$\frac{\partial f}{\partial x_j}(\mathbf{x}_0) = D_{\mathbf{e}_j} f(\mathbf{x}_0) = f'(\mathbf{x}_0) \mathbf{e}_j$$

2. If f is differentiable, then for any direction vector $\mathbf{v} = \sum_{j=1}^n v_j \mathbf{e}_j$, the directional derivative can be expanded as:

$$D_{\mathbf{v}} f(\mathbf{x}_0) = f'(\mathbf{x}_0) \mathbf{v} = f'(\mathbf{x}_0) \left(\sum_{j=1}^n v_j \mathbf{e}_j \right) = \sum_{j=1}^n v_j f'(\mathbf{x}_0) \mathbf{e}_j = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0)$$

This can be naturally represented in matrix multiplication form (where the derivative is a $m \times n$ matrix):

$$\begin{pmatrix} \left| \frac{\partial f}{\partial x_1}(\mathbf{x}_0) \right| & \left| \frac{\partial f}{\partial x_2}(\mathbf{x}_0) \right| & \cdots & \left| \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \right| \\ \left| \right| & \left| \right| & & \left| \right| \end{pmatrix}_{m \times n} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}_{n \times 1}.$$

Definition 1.4 (Frobenius Norm)

Let $A = (a_{ij})_{i,j} \in M_{m \times n}(\mathbb{R})$. We can regard A as a vector in $\mathbb{R}^{mn}$, and define its Frobenius norm by

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}.$$

Remark. The Frobenius norm can also be elegantly expressed using the trace of a matrix. For $A \in M_{m \times n}(\mathbb{R})$, consider the matrix product $AA^\top$. By the definition of matrix multiplication, the i -th diagonal entry of $AA^\top$ is given by:

$$(AA^\top)_{ii} = \sum_{j=1}^n a_{ij}(A^\top)_{ji} = \sum_{j=1}^n a_{ij}^2$$

Since the trace of a matrix is the sum of its diagonal entries, taking the trace of $AA^\top$ yields:

$$\text{tr}(AA^\top) = \sum_{i=1}^m (AA^\top)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$$

Therefore, the Frobenius norm can be equivalently written as:

$$\|A\|_F = \sqrt{\text{tr}(AA^\top)}.$$

Theorem 1.6 (Sub-multiplicativity of Frobenius Norm)

Let $A \in M_{m \times n}(\mathbb{R})$ and $B \in M_{n \times p}(\mathbb{R})$. Then the Frobenius norm satisfies the sub-multiplicative property:

$$\|AB\|_F \leq \|A\|_F \|B\|_F$$

In particular, for every column vector $\mathbf{x} \in \mathbb{R}^n$, we have $\|A\mathbf{x}\| \leq \|A\|_F \|\mathbf{x}\|$. This implies that the linear map $\mathbf{x} \mapsto A\mathbf{x}$ is continuous on $\mathbb{R}^n$.

Proof.

By the definition of matrix multiplication, the (i, j) -th entry of the product matrix AB is given by:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Applying the Cauchy-Schwarz inequality to the vectors $(a_{i1}, \dots, a_{in})$ and $(b_{1j}, \dots, b_{nj})$, we obtain an upper bound for the square of this entry:

$$|(AB)_{ij}|^2 = \left| \sum_{k=1}^n a_{ik} b_{kj} \right|^2 \leq \left(\sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{k=1}^n b_{kj}^2 \right)$$

Now, we compute the squared Frobenius norm of the matrix AB by summing over all entries ($1 \leq i \leq m$ and $1 \leq j \leq p$):

$$\begin{aligned} \|AB\|_F^2 &= \sum_{i=1}^m \sum_{j=1}^p (AB)_{ij}^2 \\ &\leq \sum_{i=1}^m \sum_{j=1}^p \left(\sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{k=1}^n b_{kj}^2 \right) \end{aligned}$$

Notice that the sum over i only depends on the terms a_{ik} , and the sum over j only depends on the terms b_{kj} . Therefore, we can separate and factor the summations:

$$\begin{aligned} \|AB\|_F^2 &\leq \left(\sum_{i=1}^m \sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{j=1}^p \sum_{k=1}^n b_{kj}^2 \right) \\ &= \|A\|_F^2 \cdot \|B\|_F^2 \end{aligned}$$

Finally, taking the square root of both sides, we get:

$$\|AB\|_F \leq \|A\|_F \|B\|_F$$

This completes the proof. ■

Theorem 1.7 (Chain Rule)

Let $E \subseteq \mathbb{R}^n$, $F \subseteq \mathbb{R}^m$, $f : E \rightarrow F$, $g : F \rightarrow \mathbb{R}^p$ and $\mathbf{x}_0 \in \text{int}(E)$. Suppose f is differentiable at $\mathbf{x}_0$, $f(\mathbf{x}_0)$ lies in the interior of F , and g is differentiable at $f(\mathbf{x}_0)$. Then $g \circ f$ is differentiable at $\mathbf{x}_0$, and

$$(g \circ f)'(\mathbf{x}_0) = g'(f(\mathbf{x}_0))f'(\mathbf{x}_0).$$

Proof.

Let $\mathbf{y}_0 = f(\mathbf{x}_0)$ and denote the derivatives by $A = f'(\mathbf{x}_0) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $B = g'(f(\mathbf{x}_0)) = g'(\mathbf{y}_0) : \mathbb{R}^m \rightarrow \mathbb{R}^p$.

Step 1: Define the error function η

Define $\eta : \mathbb{R}^m \rightarrow \mathbb{R}^p$ by:

$$\eta(\mathbf{k}) = \begin{cases} \frac{g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) - B\mathbf{k}}{\|\mathbf{k}\|} & \text{if } \mathbf{k} \neq \mathbf{0} \\ \mathbf{0} & \text{if } \mathbf{k} = \mathbf{0} \end{cases}$$

Because g is differentiable at $\mathbf{y}_0$, $\lim_{\mathbf{k} \rightarrow \mathbf{0}} \eta(\mathbf{k}) = \mathbf{0}$, which implies η is continuous at $\mathbf{0}$. Moreover, the relation

$$g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) - B\mathbf{k} = \eta(\mathbf{k})\|\mathbf{k}\|$$

is true for all $\mathbf{k} \in \mathbb{R}^m$.

Step 2: Expand the composition difference

Take $\mathbf{h}$ such that $\mathbf{x}_0 + \mathbf{h} \in E$. Let $\mathbf{k} = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)$. Consider the difference:

$$\begin{aligned} (g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) &= g(f(\mathbf{x}_0 + \mathbf{h})) - g(f(\mathbf{x}_0)) \\ &= g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) \\ &= B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) \\ &\quad + \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\| \end{aligned}$$

Subtracting $BA\mathbf{h}$ from both sides, we get:

$$\begin{aligned} (g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) - BA\mathbf{h} \\ = B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}) + \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\| \end{aligned}$$

Step 3: Estimate the first term

Note that:

$$\frac{\|B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h})\|}{\|\mathbf{h}\|} \leq \|B\| \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|}$$

By the assumption that f is differentiable at $\mathbf{x}_0$, $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|} = 0$.
By the Squeeze Theorem, the first term converges to zero.

Step 4: Estimate the second term

Since f is differentiable at $\mathbf{x}_0$, f is continuous at $\mathbf{x}_0$, meaning $\lim_{\mathbf{h} \rightarrow \mathbf{0}} (f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) = \mathbf{0}$. On the other hand, η is continuous at $\mathbf{0}$ (and $\lim_{\mathbf{k} \rightarrow \mathbf{0}} \eta(\mathbf{k}) = \mathbf{0}$), it follows that $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) = \mathbf{0}$

Next, consider:

$$\begin{aligned} \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\|}{\|\mathbf{h}\|} &= \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h} + A\mathbf{h}\|}{\|\mathbf{h}\|} \\ &\leq \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|} + \frac{\|A\mathbf{h}\|}{\|\mathbf{h}\|} \\ &\leq \underbrace{\frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|}}_{\rightarrow 0 \text{ by differentiability}} + \underbrace{\|A\|}_{\text{fixed}} \end{aligned}$$

By the Squeeze Theorem:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|\eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\| \|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\|}{\|\mathbf{h}\|} = 0$$

Combining Step 3 and Step 4, we conclude:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|(g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) - BA\mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

This proves that $g \circ f$ is differentiable at $\mathbf{x}_0$ with derivative $BA = g'(f(\mathbf{x}_0))f'(\mathbf{x}_0)$. ■

Definition 1.5

Let $E \subseteq \mathbf{R}^n$ be open and $f : E \rightarrow \mathbf{R}^m$. We say f is twice continuously differentiable on E , denoted by $f \in C^2(E)$.

1. f is continuously differentiable (its first order partial derivatives exist and are continuous).
2. For each $i = 1, 2, \dots, n$, the partial derivative $\frac{\partial f}{\partial x_i} : E \rightarrow \mathbf{R}^m$ is continuously differentiable on E (its second order partial derivatives exist and are continuous on E).

Theorem 1.8 (Clairaut)

Let E be an open set and $f \in C^2(E)$. Then for all $1 \leq i, j \leq n$ and all $\mathbf{x}_0 \in E$:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0).$$

Proof.

Let $f = (f_1, f_2, \dots, f_m)$ where $f_k : E \rightarrow \mathbb{R}$. Without loss of generality, we can assume $f : E \rightarrow \mathbb{R}$ and the point of interest is $\mathbf{x}_0 = \mathbf{0}$. (By some shifting, $F(\mathbf{x}) = f(\mathbf{x} + \mathbf{x}_0) \Rightarrow F(\mathbf{0}) = f(\mathbf{x}_0)$). We also assume $i \neq j$.

Since $f \in C^2(E)$ and $\mathbf{0} \in E$, given any $\epsilon > 0$, there exists $\delta_0 > 0$ such that:

$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2} \quad \text{and} \quad \left| \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) - \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

if $\|\mathbf{x}\| < 2\delta_0$.

Step 1: The Second-Order Difference (DiD Approximation)

Fix $\delta < \delta_0$, and consider the rectangle. Define the second-order difference X :

$$X = f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_i) - f(\delta \mathbf{e}_j) + f(\mathbf{0})$$

We can group the terms in two different ways:

$$X = (f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_i)) - (f(\delta \mathbf{e}_j) - f(\mathbf{0})) \quad (1)$$

$$X = (f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_j)) - (f(\delta \mathbf{e}_i) - f(\mathbf{0})) \quad (2)$$

Step 2: Apply Mean Value Theorem for Grouping (1)

Let $\phi(t) = f(t\mathbf{e}_i + \delta \mathbf{e}_j) - f(t\mathbf{e}_i)$. Then from (1), we have:

$$X = \phi(\delta) - \phi(0)$$

By the MVT, $X = \delta \phi'(c_i)$ for some $c_i \in (0, \delta)$ (*).

Note that $\phi'(t) = \frac{\partial f}{\partial x_i}(t\mathbf{e}_i + \delta \mathbf{e}_j) - \frac{\partial f}{\partial x_i}(t\mathbf{e}_i)$. So, $\phi'(c_i) = \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i + \delta \mathbf{e}_j) - \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i)$.

Now, let $g(s) = \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i + s\mathbf{e}_j)$. Then $g'(s) = \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + s\mathbf{e}_j)$. We can rewrite (*) as:

$$\phi'(c_i) = g(\delta) - g(0)$$

By applying MVT again to $g(s)$, there exists $c_j \in (0, \delta)$ such that $g(\delta) - g(0) = \delta g'(c_j)$. Thus, $\phi'(c_i) = \delta \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j)$ (**).

From (*) and (**), we obtain:

$$X = \delta^2 \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j) \quad \text{where } 0 < c_i < \delta, 0 < c_j < \delta.$$

Step 3: Symmetric Argument for Grouping (2)

Similarly, starting from (2) and taking derivatives first with respect to x_j and then x_i , MVT gives us:

$$X = \delta^2 \frac{\partial^2 f}{\partial x_i \partial x_j}(d_i \mathbf{e}_i + d_j \mathbf{e}_j) \quad \text{where } 0 < d_i < \delta, 0 < d_j < \delta.$$

Step 4: Bounding the Error using Continuity

Notice that the norm of the points we found is strictly bounded:

$$\|c_i \mathbf{e}_i + c_j \mathbf{e}_j\| = \sqrt{c_i^2 + c_j^2} < \sqrt{\delta^2 + \delta^2} = \sqrt{2}\delta < \sqrt{2}\delta_0 < 2\delta_0$$

Because these points fall within the $2\delta_0$ radius, our initial continuity conditions apply:

$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2} \implies \left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

Similarly,

$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

By the triangle inequality:

$$\left| \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \epsilon \quad \forall \epsilon > 0$$

Hence, we conclude that:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}).$$

■

Remark (Numerical Approximation via Second-Order Difference).

From the proof above, we know that by the continuity of $\frac{\partial^2 f}{\partial x_j \partial x_i}$ and $\frac{\partial^2 f}{\partial x_i \partial x_j}$,

$$\lim_{\delta \rightarrow 0} \frac{X}{\delta^2} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \quad \text{and} \quad \lim_{\delta \rightarrow 0} \frac{X}{\delta^2} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0})$$

Hence, if $f \in C^2(E)$, for any point $\mathbf{x}_0$, we can use numerical approximation by a second-order finite difference without computing the derivative directly:

$$\lim_{\delta \rightarrow 0} \frac{f(\mathbf{x}_0 + \delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\mathbf{x}_0 + \delta \mathbf{e}_i) - f(\mathbf{x}_0 + \delta \mathbf{e}_j) + f(\mathbf{x}_0)}{\delta^2} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0)$$

1.2 Inverse Function Theorem

Definition 1.6 (Metric Space)

The pair (X, d) is called a metric space if a set X endowed with the function $d : X \times X \rightarrow \mathbb{R}$ such that ...

Definition 1.7 (Convergence)

Given a sequence $\{x_n\}_{n=1}^\infty$ in X . We say the sequence converges to x if $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. for $n \geq N$, $d(x_n, x) < \epsilon$.

Definition 1.8 (Cauchy Sequence)

A sequence $\{x_n\}_{n=1}^{\infty}$ is said to be *Cauchy* if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. for $m, n \geq N, d(x_n, x_m) < \varepsilon$.

Definition 1.9 (Completeness)

(X, d) is *complete* if every Cauchy sequence in X converges to a limit in X .

Theorem 1.9

Let E be a closed subset in X , and (X, d) is a complete metric space. Then (E, d) is also complete.

Proof.

Take any Cauchy sequence in E , which is also a Cauchy in X . By completeness of X , the sequence converges to some point in X . Since a closed set contains all limit point, the Cauchy sequence converges to a limit in E . ■

Definition 1.10 (Contraction)

Let (X, d) be a metric space and $T : X \rightarrow X$ be a map.

1. We say T is a *contraction* if $d(T(x), T(y)) \leq d(x, y) \forall x, y \in X$.
2. T is a *strict contraction* if there exists a constant $0 < c < 1$ s.t. $d(T(x), T(y)) \leq cd(x, y)$.

Remark. Under the standard setting, the term **contraction** simply refers to a **strict contraction**. Terence Tao use “contraction” for maps where $d(T(x), T(y)) \leq d(x, y)$ and explicitly write “strict contraction” when the constant is strictly less than 1.

Definition 1.11 (Fixed Point)

Let $T : X \rightarrow X$ be a map and $x^* \in X$. We say x^* is a *fixed point* of T if $T(x^*) = x^*$.

Remark. Suppose X is a subset of $\mathbb{R}$. The existence of a fixed point means that the graph of $y = f(x)$ intersect with $y = x$.

Theorem 1.10 (Contraction Mapping Theorem)

Let (X, d) be a metric space and $T : X \rightarrow X$ be a (strict) contraction. Then T has at most one fixed point. Moreover, if we assume X is non-empty and complete, then T has exactly one fixed point.

Proof.

Suppose x, y are distinct fixed point of T , consider $d(x, y) = d(T(x), T(y)) \leq cd(x, y)$ for some $c \in (0, 1)$. The inequality $(1 - c)d(x, y) \leq 0$ forces $x = y$.

Now we further assume X is complete. Take any point x_0 , construct the sequence

$\{x_n\}_{n=0}^\infty$, $x_{n+1} = T(x_n) \forall n \in \mathbb{N}$. Observe that $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$ for $n \geq 1$. Suppose $n > m \geq 1$,

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n-1}) + d(x_{n-1}, x_{n-2}) + \cdots + d(x_{m+1}, x_m) \\ &\leq c^{n-1} d(x_1, x_0) + c^{n-2} d(x_1, x_0) + \cdots + c^m d(x_1, x_0) \\ &\leq \frac{c^m}{1-c} d(x_1, x_0). \end{aligned}$$

Take $m \rightarrow \infty$, $c^m \rightarrow 0$, which shows $\{x_n\}_{n=1}^\infty$ is a Cauchy in X . Since X is complete, $\lim_{n \rightarrow \infty} x_n$ exists and converges to x^* . Finally,

$$\begin{aligned} d(x^*, T(x^*)) &\leq d(x^*, x_{n+1}) + d(x_{n+1}, T(x^*)) \\ &\leq d(x^*, x_{n+1}) + d(T(x_n), T(x^*)) \\ &\leq d(x^*, x_{n+1}) + cd(x_n, x^*). \end{aligned}$$

This shows $x^* = T(x^*)$. ■

Exercise 1.1

Show that the contraction $T : X \rightarrow X$ be a continuous function.

Lemma

Let $B(\mathbf{0}, r)$ be an open ball in $\mathbb{R}^n$ and $g : B(\mathbf{0}, r) \rightarrow \mathbb{R}^n$ be a map with $g(\mathbf{0}) = \mathbf{0}$. If $\|g(\mathbf{x}) - g(\mathbf{y})\| \leq \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|$ for $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$, then the function $f : B(\mathbf{0}, r) \rightarrow \mathbb{R}^n$ defined by $f(\mathbf{x}) = \mathbf{x} + g(\mathbf{x})$ is one-to-one, and the image of $f(B(\mathbf{0}, r))$ contains $B(\mathbf{0}, \frac{r}{2})$.

Proof.

We first show injection. Suppose $f(\mathbf{x}) = f(\mathbf{y})$, by construction, we have $\|\mathbf{x} - \mathbf{y}\| = \|g(\mathbf{x}) - g(\mathbf{y})\| \leq \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|$. The inequality $\frac{1}{2}\|\mathbf{x} - \mathbf{y}\| \leq 0$ forces $\|\mathbf{x} - \mathbf{y}\| = 0$. So f is one-to-one.

Next, we want to show $B(\mathbf{0}, \frac{r}{2}) \subseteq f(B(\mathbf{0}, r))$. It suffices to show $\forall \mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$, $\exists \mathbf{x} \in B(\mathbf{0}, r)$ s.t. $f(\mathbf{x}) = \mathbf{y}$. Define $F(\mathbf{x}) = \mathbf{y} - g(\mathbf{x})$. If we can find the fixed point of F , then we equivalently find $\mathbf{y} = f(\mathbf{x})$. Fix $\mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$. $\|\mathbf{y}\| < \frac{r}{2}$. $\exists \varepsilon > 0$ s.t. $\|\mathbf{y}\| \leq \frac{r-\varepsilon}{2}$. Consider the map $F : \overline{B(\mathbf{0}, r-\varepsilon)} \rightarrow \overline{B(\mathbf{0}, r-\varepsilon)}$. Take $\mathbf{x} \in \overline{B(\mathbf{0}, r-\varepsilon)}$, $\|\mathbf{x}\| \leq r-\varepsilon$. Then we have $\|F(\mathbf{x})\| \leq \|\mathbf{y}\| + \|g(\mathbf{x})\| \leq \frac{r-\varepsilon}{2} + \|g(\mathbf{x})\| \leq \frac{r-\varepsilon}{2} + \frac{r-\varepsilon}{2} = r-\varepsilon$, where $\|g(\mathbf{x})\| = \|g(\mathbf{x}) - g(\mathbf{0})\| \leq \frac{1}{2}\|\mathbf{x}\| \leq \frac{1}{2}(r-\varepsilon)$. So $F(\mathbf{x}) \in \overline{B(\mathbf{0}, r-\varepsilon)}$. Given a fixed $\mathbf{y}$, $\|F(\mathbf{x}_1) - F(\mathbf{x}_2)\| = \|\mathbf{y} - g(\mathbf{x}_1) - (\mathbf{y} - g(\mathbf{x}_2))\| \leq \|g(\mathbf{x}_2) - g(\mathbf{x}_1)\| \leq \frac{1}{2}\|\mathbf{x}_2 - \mathbf{x}_1\|$. Hence, $F : \overline{B(\mathbf{0}, r-\varepsilon)} \rightarrow \overline{B(\mathbf{0}, r-\varepsilon)}$ is a contraction $\forall \mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$. Since $\overline{B(\mathbf{0}, r-\varepsilon)}$ is complete, $\exists \mathbf{x}^* \in \overline{B(\mathbf{0}, r-\varepsilon)}$ s.t. $\mathbf{x}^* = F(\mathbf{x}^*)$. This shows $F(\mathbf{x}^*) = \mathbf{y} - g(\mathbf{x}^*) = \mathbf{x}^*$, and hence $\mathbf{x}^* + g(\mathbf{x}^*) = \mathbf{y}$. ■

Theorem 1.11 (Inverse Function Theorem)

Let E be open in $\mathbb{R}^n$ and $f : E \rightarrow \mathbb{R}^n \in C^1(E)$. Suppose the linear transformation $f'(\mathbf{x}_0) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible, then there exists an open set $U \subseteq E$ containing $\mathbf{x}_0$ and an open set $V \subseteq \mathbb{R}^n$ containing $f(\mathbf{x}_0)$ such that $f : U \rightarrow V$ is a bijection. Moreover, the inverse map $f^{-1} : V \rightarrow U$ is also continuously differentiable, and

$$(f^{-1})'(f(\mathbf{x}_0)) = (f'(\mathbf{x}_0))^{-1}.$$

Proof.

Let $A = f'(\mathbf{x}_0)$. Firstly, f is differentiable at $\mathbf{x}_0$, so:

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x}\|}{\|\mathbf{x}\|} = 0.$$

Given A is invertible, as $\mathbf{x} \rightarrow \mathbf{0}$, we have:

$$\frac{\|A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x})\|}{\|\mathbf{x}\|} \leq \|A^{-1}\| \frac{\|f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x}\|}{\|\mathbf{x}\|} \rightarrow 0.$$

Step 1: Construct the auxiliary function g and show it is a contraction

Let $g(\mathbf{x}) = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0)) - \mathbf{x}$. It is clear that $g(\mathbf{0}) = \mathbf{0}$. Substituting this into our limit condition yields $\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} = 0$. Rewrite the limit condition as:

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|g(\mathbf{x}) - g(\mathbf{0}) - g'(\mathbf{0})\mathbf{x}\|}{\|\mathbf{x}\|} = 0,$$

which shows that g is differentiable at $\mathbf{0}$ with $g'(\mathbf{0}) = \mathbf{0}$.

Since both f and A are C^1 , g is also C^1 . By definition, there exists an $r > 0$ such that for all $\mathbf{x} \in B(\mathbf{0}, r)$:

$$\left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) - \frac{\partial g}{\partial x_j}(\mathbf{0}) \right\| = \left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \frac{1}{2n^2}.$$

For $\mathbf{x} \in B(\mathbf{0}, r)$ and $\mathbf{v} \in \mathbb{R}^n$,

$$\|D_{\mathbf{v}}g(\mathbf{x})\| = \left\| \sum_{j=1}^n v_j \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \sum_{j=1}^n |v_j| \left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \sum_{j=1}^n \|\mathbf{v}\| \frac{1}{2n^2} = \frac{1}{2n} \|\mathbf{v}\| \leq \frac{1}{2} \|\mathbf{v}\|.$$

For $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$, consider:

$$\begin{aligned} g(\mathbf{y}) - g(\mathbf{x}) &= \int_0^1 \frac{d}{dt} g(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) dt \quad (\text{FTC(1)}) \\ &= \int_0^1 Dg(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) dt \quad (\text{Chain rule}) \\ &= \int_0^1 D_{\mathbf{y}-\mathbf{x}}g(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) dt \quad (\text{Since } g \text{ is } C^1) \end{aligned}$$

Therefore,

$$\|g(\mathbf{y}) - g(\mathbf{x})\| = \left\| \int_0^1 D_{\mathbf{y}-\mathbf{x}}g(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))dt \right\| \leq \int_0^1 \frac{1}{2}\|\mathbf{y} - \mathbf{x}\|dt \leq \frac{1}{2}\|\mathbf{y} - \mathbf{x}\|,$$

which means g is a contraction.

Step 2: Define Φ , establish local bijection, and show Φ^{-1} is continuous

Define $\Phi(\mathbf{x}) = g(\mathbf{x}) + \mathbf{x} = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0))$, where $g(\mathbf{0}) = \mathbf{0}$ and g is a contraction. Apply the lemma for the Inverse Function Theorem, Φ is one-to-one on $B(\mathbf{0}, r)$ and $B(\mathbf{0}, \frac{r}{2}) \subseteq \Phi(B(\mathbf{0}, r))$. Set $W = \Phi^{-1}(B(\mathbf{0}, \frac{r}{2}))$, then $\Phi|_W : W \rightarrow B(\mathbf{0}, \frac{r}{2})$ is a bijection.

Clearly, Φ is continuous. We claim that Φ^{-1} is also continuous. For $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$:

$$\|\Phi(\mathbf{x}) - \Phi(\mathbf{y})\| = \|(\mathbf{x} - \mathbf{y}) + (g(\mathbf{x}) - g(\mathbf{y}))\| \geq \|\mathbf{x} - \mathbf{y}\| - \|g(\mathbf{x}) - g(\mathbf{y})\| \geq \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|.$$

Let $\mathbf{x} = \Phi^{-1}(\mathbf{u})$ and $\mathbf{y} = \Phi^{-1}(\mathbf{v})$, with $\mathbf{u}, \mathbf{v} \in B(\mathbf{0}, \frac{r}{2}) \subseteq \Phi(B(\mathbf{0}, r))$, then:

$$\begin{aligned} \|\Phi(\Phi^{-1}(\mathbf{u})) - \Phi(\Phi^{-1}(\mathbf{v}))\| &\geq \frac{1}{2}\|\Phi^{-1}(\mathbf{u}) - \Phi^{-1}(\mathbf{v})\| \\ \|\Phi^{-1}(\mathbf{u}) - \Phi^{-1}(\mathbf{v})\| &\leq 2\|\mathbf{u} - \mathbf{v}\|. \end{aligned}$$

This shows that Φ^{-1} is Lipschitz continuous and hence continuous.

Step 3: Prove Φ^{-1} is differentiable at $\mathbf{0}$

We claim that the derivative of Φ^{-1} at $\mathbf{0}$ is the identity map id . Let $\mathbf{z} \in B(\mathbf{0}, \frac{r}{2}) \setminus \{\mathbf{0}\}$.

Since $\Phi|_W$ is bijective, let $\mathbf{z} = \Phi(\mathbf{x}) = \mathbf{x} + g(\mathbf{x})$, which means $\mathbf{x} = \Phi^{-1}(\mathbf{z})$ and $\mathbf{x} \neq \mathbf{0}$.

We want to show:

$$\lim_{\mathbf{z} \rightarrow \mathbf{0}} \frac{\|\Phi^{-1}(\mathbf{z}) - \Phi^{-1}(\mathbf{0}) - \text{id}(\mathbf{z})\|}{\|\mathbf{z}\|} = 0.$$

Since $\Phi^{-1}(\mathbf{0}) = \mathbf{0}$, the expression inside the limit simplifies to:

$$\frac{\|\Phi^{-1}(\mathbf{z}) - \mathbf{z}\|}{\|\mathbf{z}\|} = \frac{\|\mathbf{x} - (\mathbf{x} + g(\mathbf{x}))\|}{\|\mathbf{z}\|} = \frac{\| -g(\mathbf{x}) \|}{\|\mathbf{z}\|} = \frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot \frac{\|\mathbf{x}\|}{\|\mathbf{z}\|}.$$

From the Lipschitz continuity in Step 2, taking $\mathbf{v} = \mathbf{0}$, we have $\|\Phi^{-1}(\mathbf{z})\| \leq 2\|\mathbf{z}\|$, which implies $\frac{\|\mathbf{x}\|}{\|\mathbf{z}\|} \leq 2$. Furthermore, as $\mathbf{z} \rightarrow \mathbf{0}$, sequential continuity dictates that $\mathbf{x} = \Phi^{-1}(\mathbf{z}) \rightarrow \mathbf{0}$. Using the Squeeze Theorem:

$$\lim_{\mathbf{z} \rightarrow \mathbf{0}} \frac{\|\Phi^{-1}(\mathbf{z}) - \mathbf{z}\|}{\|\mathbf{z}\|} = \lim_{\mathbf{x} \rightarrow \mathbf{0}} \left(\frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot \frac{\|\mathbf{x}\|}{\|\mathbf{z}\|} \right) \leq \lim_{\mathbf{x} \rightarrow \mathbf{0}} \left(\frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot 2 \right) = 0 \cdot 2 = 0.$$

This proves that Φ^{-1} is differentiable at $\mathbf{0}$ with $(\Phi^{-1})'(\mathbf{0}) = \text{id}$.

Step 4: Prove f^{-1} is differentiable at $f(\mathbf{x}_0)$

Recall the definition $\Phi(\mathbf{x}) = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0))$. Multiplying by A and rearranging yields:

$$f(\mathbf{x} + \mathbf{x}_0) = A\Phi(\mathbf{x}) + f(\mathbf{x}_0).$$

Since Φ is a local bijection, f is also one-to-one and onto near $\mathbf{x}_0$. Let $\mathbf{y} = f(\mathbf{x} + \mathbf{x}_0)$, then $\mathbf{y} = A\Phi(\mathbf{x}) + f(\mathbf{x}_0)$. We can express $\mathbf{x}$ as:

$$\mathbf{x} = \Phi^{-1}(A^{-1}(\mathbf{y} - f(\mathbf{x}_0))).$$

Thus, the inverse function f^{-1} is given by:

$$f^{-1}(\mathbf{y}) = \mathbf{x} + \mathbf{x}_0 = \Phi^{-1}(A^{-1}(\mathbf{y} - f(\mathbf{x}_0))) + \mathbf{x}_0.$$

We apply the Chain Rule to differentiate f^{-1} at $f(\mathbf{x}_0)$:

$$\begin{aligned}(f^{-1})'(f(\mathbf{x}_0)) &= (\Phi^{-1})'(A^{-1}(f(\mathbf{x}_0) - f(\mathbf{x}_0))) \circ (A^{-1})'(f(\mathbf{x}_0) - f(\mathbf{x}_0)) \\ &= (\Phi^{-1})'(\mathbf{0}) \circ A^{-1}(\mathbf{0}) \\ &= \text{id} \circ A^{-1} = A^{-1} = (f'(\mathbf{x}_0))^{-1}.\end{aligned}$$

■